

DEFINITION Sequence

A **sequence** $\{a_n\}$ is an ordered list of numbers of the form

$$\{a_1, a_2, a_3, \dots, a_n, \dots\}.$$

A sequence may be generated by a **recurrence relation** of the form $a_{n+1} = f(a_n)$, for $n = 1, 2, 3, \dots$, where a_1 is given. A sequence may also be defined with an **explicit formula** of the form $a_n = f(n)$, for $n = 1, 2, 3, \dots$.

EXAMPLE 1 Explicit formulas Use the explicit formula for $\{a_n\}_{n=1}^{\infty}$ to write the first four terms of each sequence. Sketch a graph of the sequence.

a. $a_n = \frac{1}{2^n}$ b. $a_n = \frac{(-1)^n n}{n^2 + 1}$

```
from matplotlib import pyplot as plt
```

```
def f(x):
    return 1/2**x
```

```
def plot_sequence(series, n):
    x = [i for i in range(1, n+1)]
    y = []
```

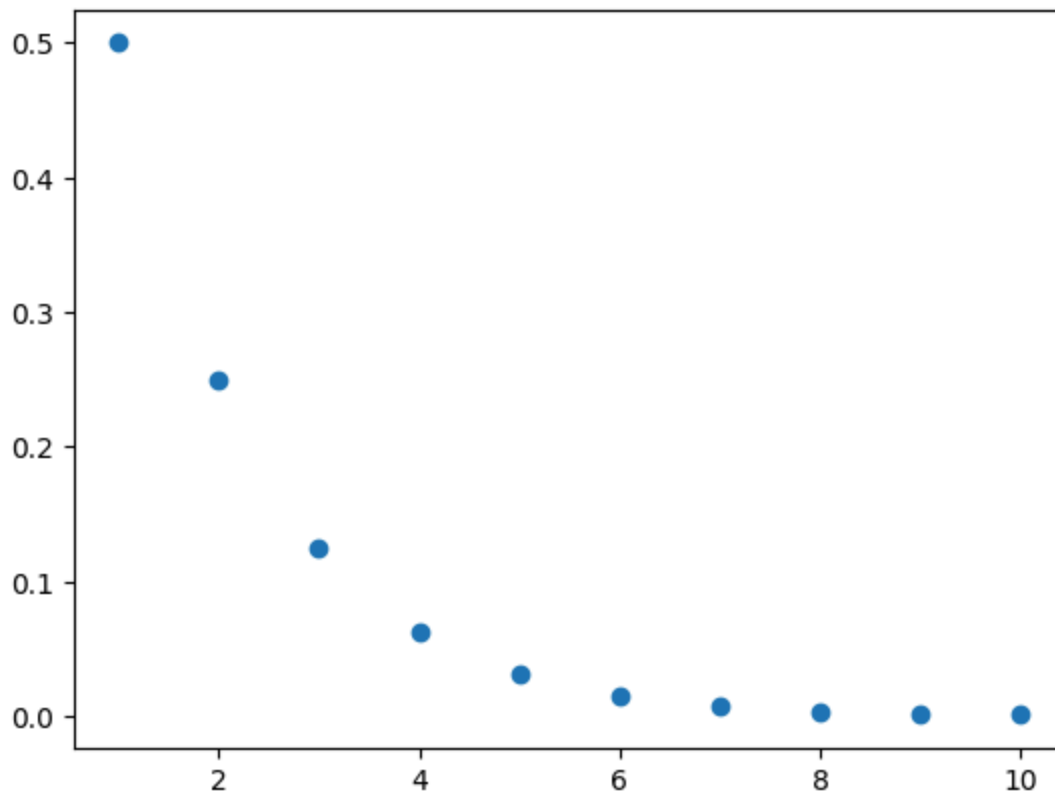
```
    for i in x:
        term = series(i)
        y.append(term)
        print(term)
```

```
    print()
    plt.scatter(x,y)
```

```
plot_sequence(f,10)
```



0.5
0.25
0.125
0.0625
0.03125
0.015625
0.0078125
0.00390625
0.001953125
0.0009765625



```
def g(x):  
    return ((-1)**x*x)/(x**2+1)
```

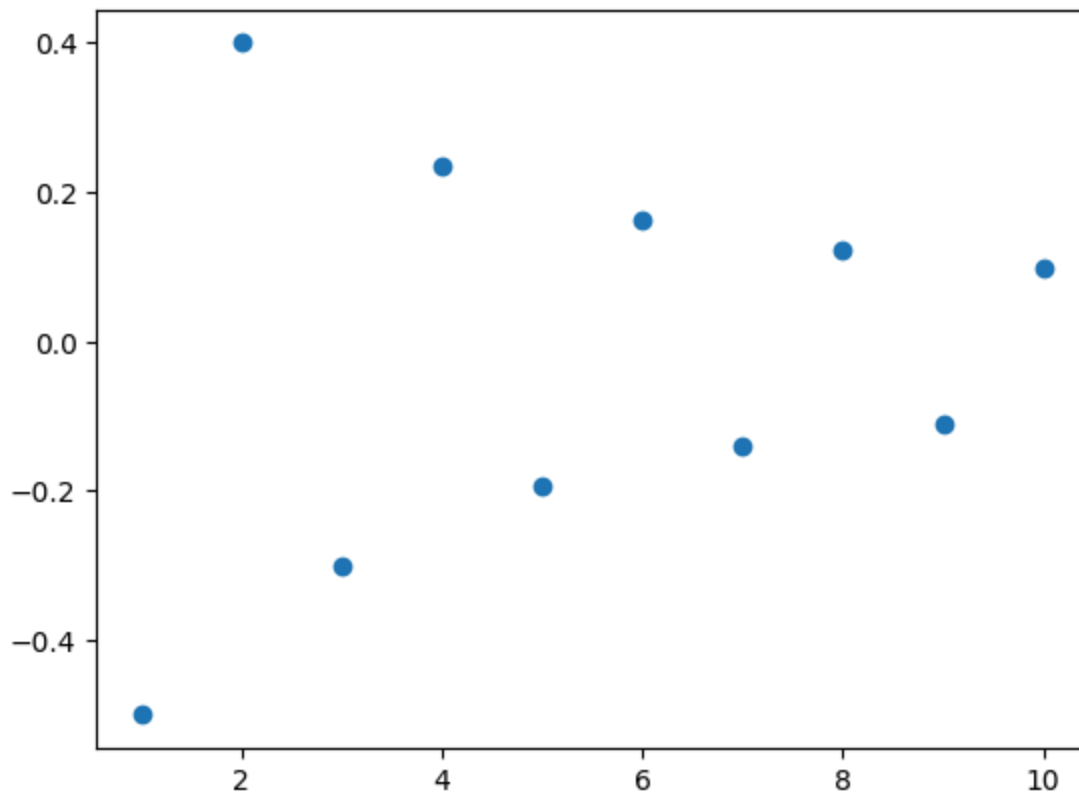
```
plot_sequence(g,10)
```



```

-0.5
0.4
-0.3
0.23529411764705882
-0.19230769230769232
0.16216216216216217
-0.14
0.12307692307692308
-0.10975609756097561
0.09900990099009901

```

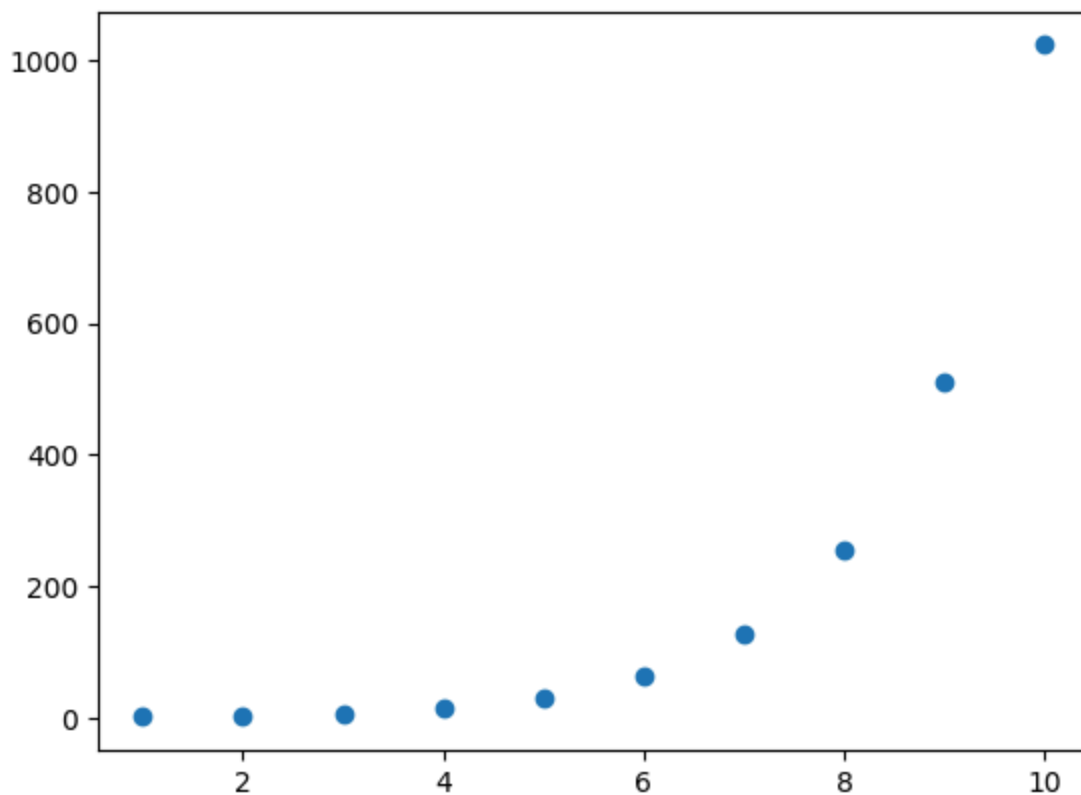


EXAMPLE 2 Recurrence relations Use the recurrence relation for $\{a_n\}_{n=1}^{\infty}$ to write the first four terms of the sequences

$$a_{n+1} = 2a_n + 1, a_1 = 1 \quad \text{and} \quad a_{n+1} = 2a_n + 1, a_1 = -1.$$

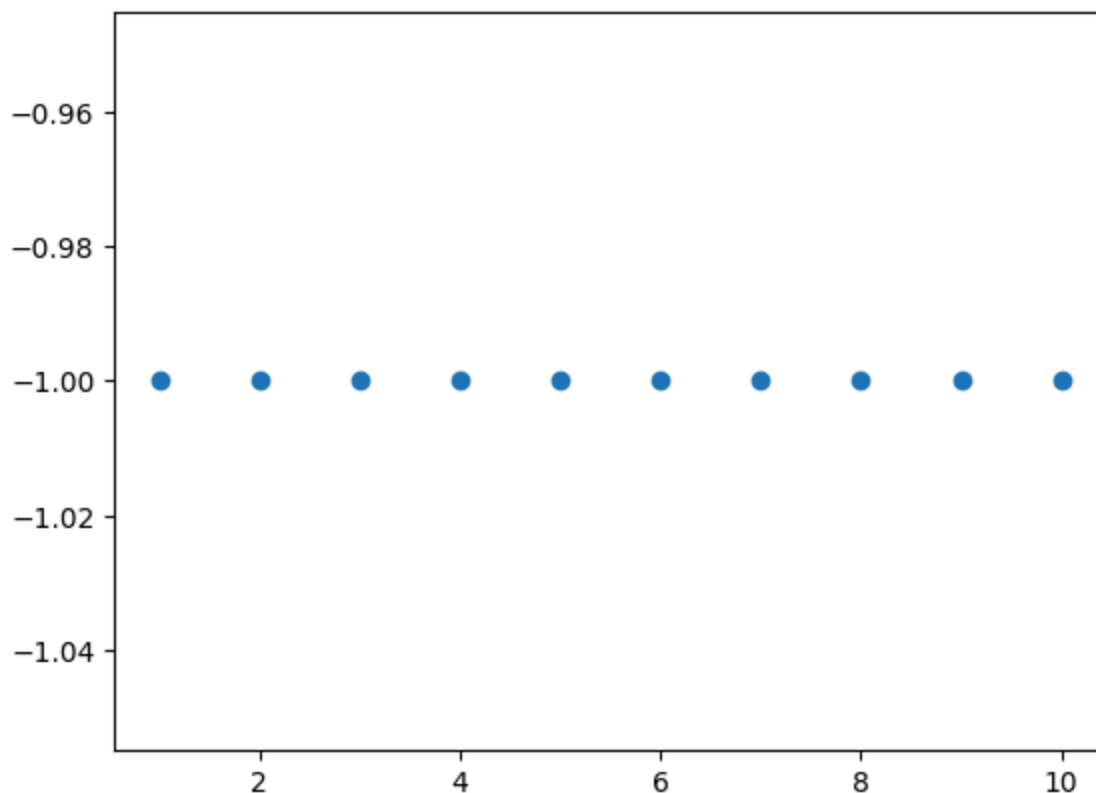
```
def f(x):  
    return 2 * x + 1  
  
def implicit_plot(f,a,n):  
    a = a  
    print(a)  
    x = [i for i in range(1,n+1)]  
    y = [a]  
    for i in range(1,n):  
        term = f(a)  
        y.append(term)  
        print(term)  
        a = term  
  
    print()  
    plt.scatter(x,y)  
  
implicit_plot(f, 1, 10)
```

```
⇒ 1  
3  
7  
15  
31  
63  
127  
255  
511  
1023
```



```
def g(x):  
    return 2 * x + 1  
  
implicit_plot(g,-1,10)
```

⇒ -1
-1
-1
-1
-1
-1
-1
-1
-1
-1
-1



DEFINITION Limit of a Sequence

If the terms of a sequence $\{a_n\}$ approach a unique number L as n increases—that is, if a_n can be made arbitrarily close to L by taking n sufficiently large—then we say $\lim_{n \rightarrow \infty} a_n = L$ exists, and the sequence **converges** to L . If the terms of the sequence do not approach a single number as n increases, the sequence has no limit, and the sequence **diverges**.

EXAMPLE 5 Limit of a sequence Enumerate and graph the terms of the following sequence, and make a conjecture about its limit.

$$a_n = \frac{4n^3}{n^3 + 1}, \quad \text{for } n = 1, 2, 3, \dots \quad \text{Explicit formula}$$

```
def f(x):
    return (4*x**3)/(x**3+1)
```

```
plot_sequence(f, 10)
```

```
↔ 2.0
3.5555555555555554
3.857142857142857
3.9384615384615387
3.9682539682539684
3.9815668202764978
3.988372093023256
3.992202729044834
3.9945205479452053
3.996003996003996
```

